# Lecture 19
# Normal Probabilities, the Central Limit Theorem and Confidence Intervals

# Review

**1 What is a sampling distribution?**

The distribution of a statistic, such as $\bar{x}$, over all possible samples of size n.

**2 What is the standard error of $\bar{x}$?**

$\sigma / \sqrt{n}$: how much $\bar{x}$ varies from sample to sample.

**3 For a continuous random variable, what is P(X = c)?**

0: a single value has no width, so it has no area under the curve.

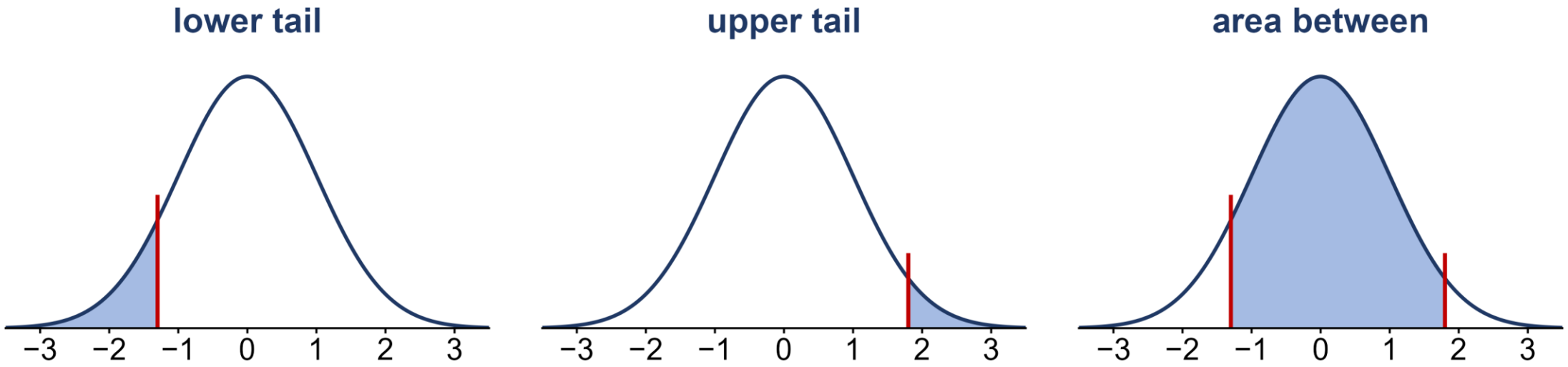
**4 What is the standard normal distribution?**

N(0, 1): the normal distribution of z-scores, with mean 0 and standard deviation 1.

# Three kinds of area

The z-table gives one kind of area only: the area to the left of z, $P(Z \le z)$.

**How would you get each shaded area from the z-table?**



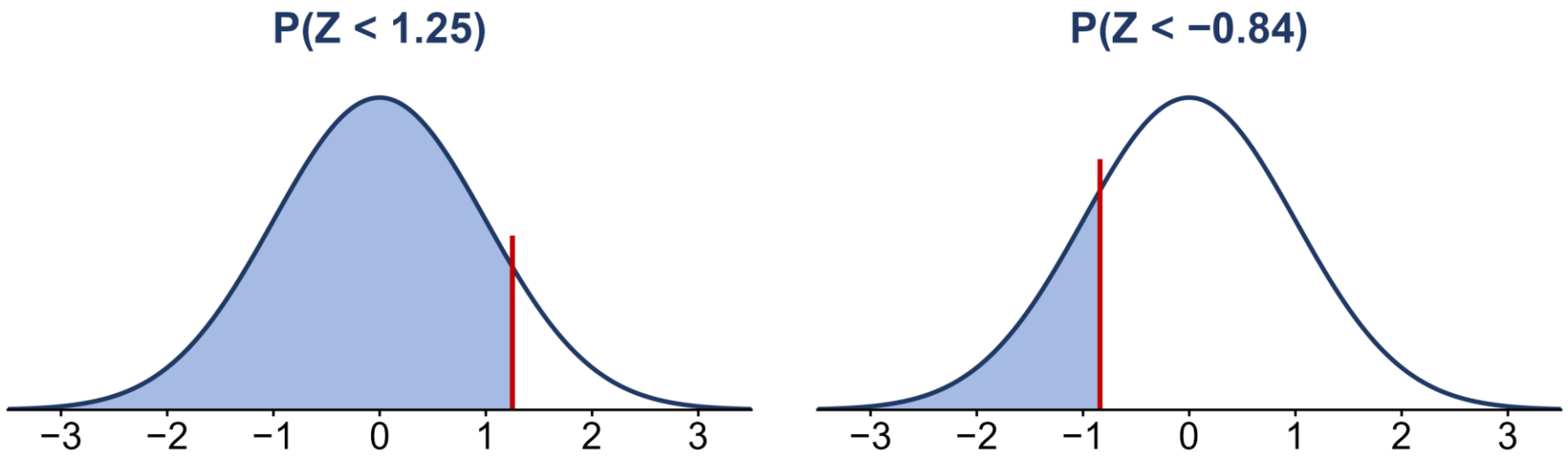

**lower tail: $P(Z < z)$ = table value** **upper tail: $P(Z > z) = 1 -$ table value**

**area between: $P(a < Z < b)$ = table value at b − table value at a**

# Example: lower tail

Z ~ N(0, 1), the standard normal. Find each shaded area with the z-table.

**(a) P(Z < 1.25)?** **(b) P(Z < −0.84)?**

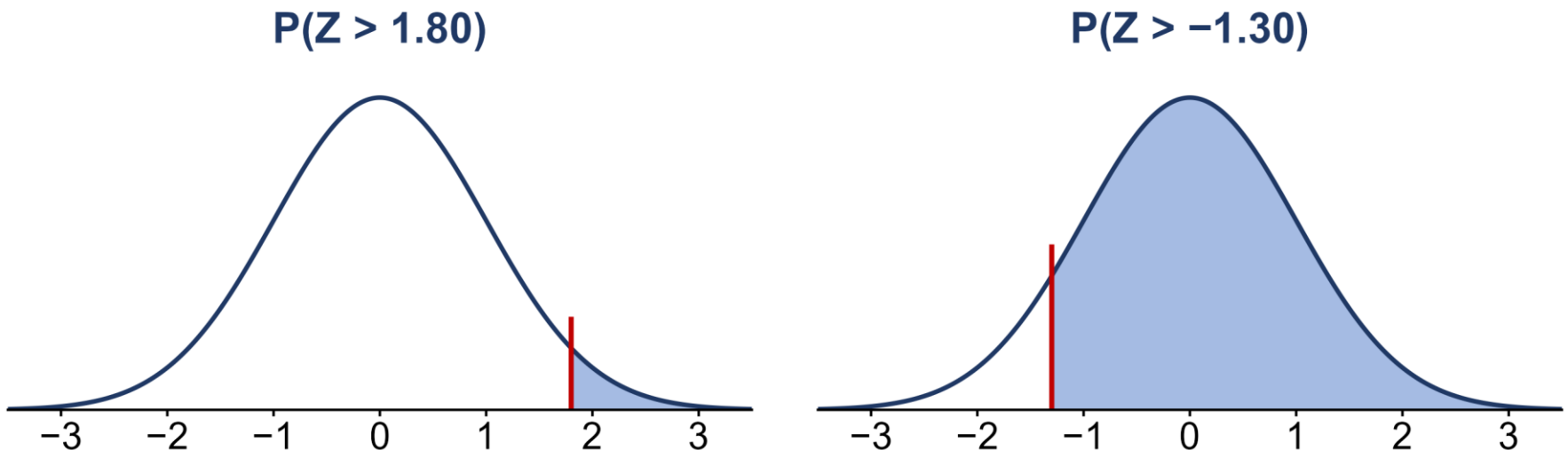



**(a) 0.8944** **(b) 0.2005**

# Example: upper tail

Z ~ N(0, 1). The table gives the area to the left, so subtract it from 1.

**(a) P(Z > 1.80)?** **(b) P(Z > −1.30)?**

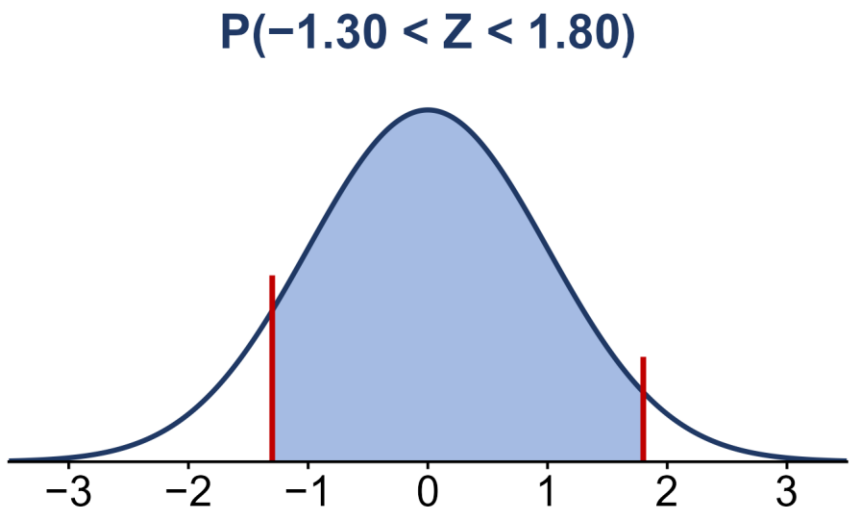


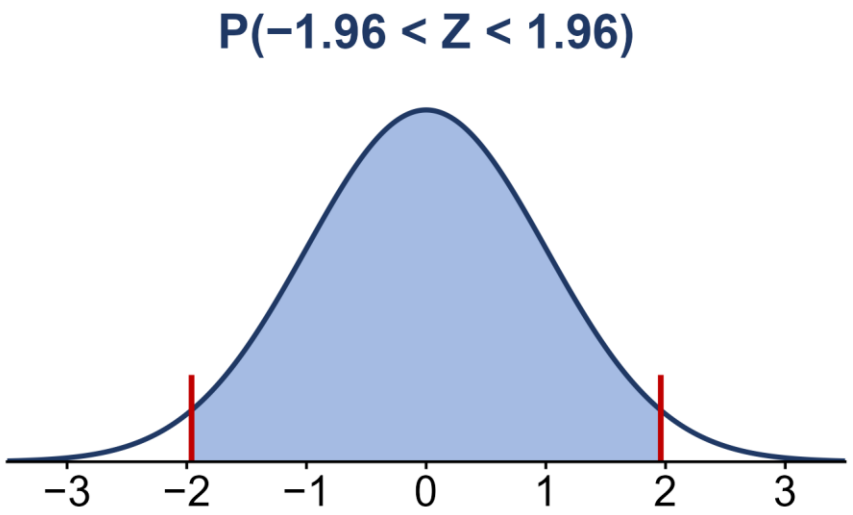

**(a) 1 − 0.9641 = 0.0359** **(b) 1 − 0.0968 = 0.9032**

# Example: area between

Z ~ N(0, 1). Take the area to the left of the larger value, minus the area to the left of the smaller.

**(a) P(−1.30 < Z < 1.80)?** **(b) P(−1.96 < Z < 1.96)?**




**(a) 0.9641 − 0.0968 = 0.8673** **(b) 0.9750 − 0.0250 = 0.9500**

# Practice: finding areas

Sketch the curve and shade the area first, then use the z-table.

**$Z \sim N(0, 1)$**

**(a) $P(Z \leq 2.00)$?  (b) $P(Z > 0.50)$?**

**(c) $P(-1 < Z < 1)$?  (d) $P(-2 < Z < 2)$?**

**Which rule from Week 6 do (c) and (d) match?**

**(a) 0.9772  (b) 1 − 0.6915 = 0.3085**

**(c) 0.8413 − 0.1587 = 0.6826  (d) 0.9772 − 0.0228 = 0.9544**

**The empirical rule: about 68% within 1 SD and 95% within 2 SD**

# Data: waiting for a goal

The 2022 World Cup had 172 goals in 64 matches: 2.69 goals per 90-minute match.

**X = minutes until the next goal, simulated at 2.69 goals per 90 minutes**

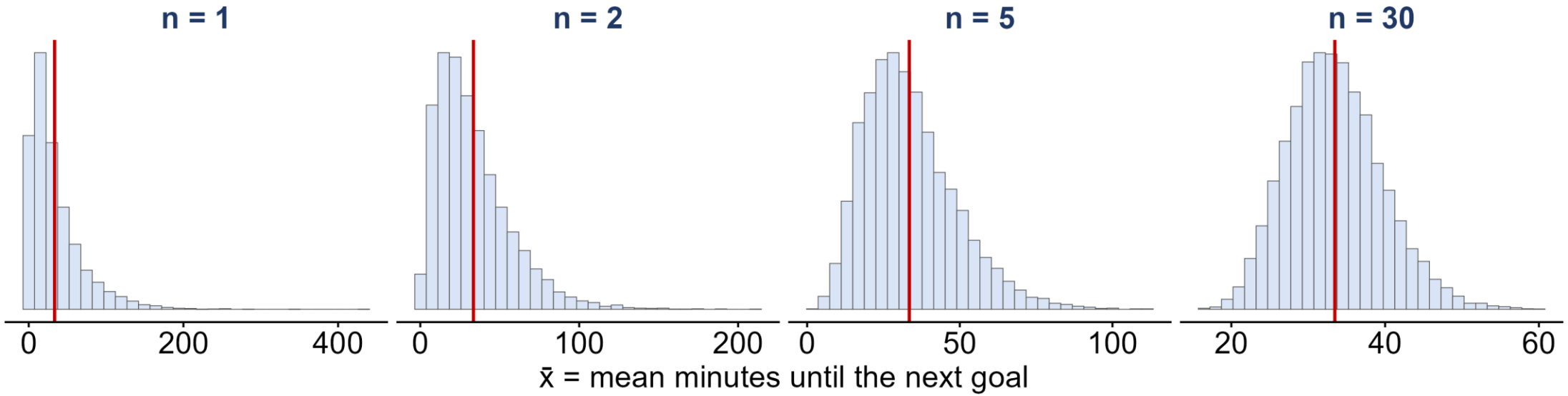
| WAIT | MINUTES |
|---|---|
| 1 | 1.6 |
| 2 | 54.3 |
| 3 | 57.3 |
| 4 | 27.2 |
| 5 | 18.3 |
| … | … |

$\mu = 90 / 2.69 \approx 33.5$ min; $\sigma \approx 33.5$ min. Most waits are short and a few are very long.

# Averaging a skewed variable

Each panel: draw n waits, average them to get $\bar{x}$, and repeat 10,000 times (n = 1 is a single wait).

**As n grows, what happens to the shape, the center and the spread of $\bar{x}$?**




**Shape: the skew fades into a bell. Center: stays at 33.5. Spread: shrinks like σ / √n**

# The central limit theorem

central limit theorem

**for large n, $\bar{x}$ is approximately $N(\mu, \sigma / \sqrt{n})$, whatever the shape of the population**

how large is large enough?

**$n \geq 30$ for $\bar{x}$;  $np > 5$ and $n(1 - p) > 5$ for $\hat{p}$**

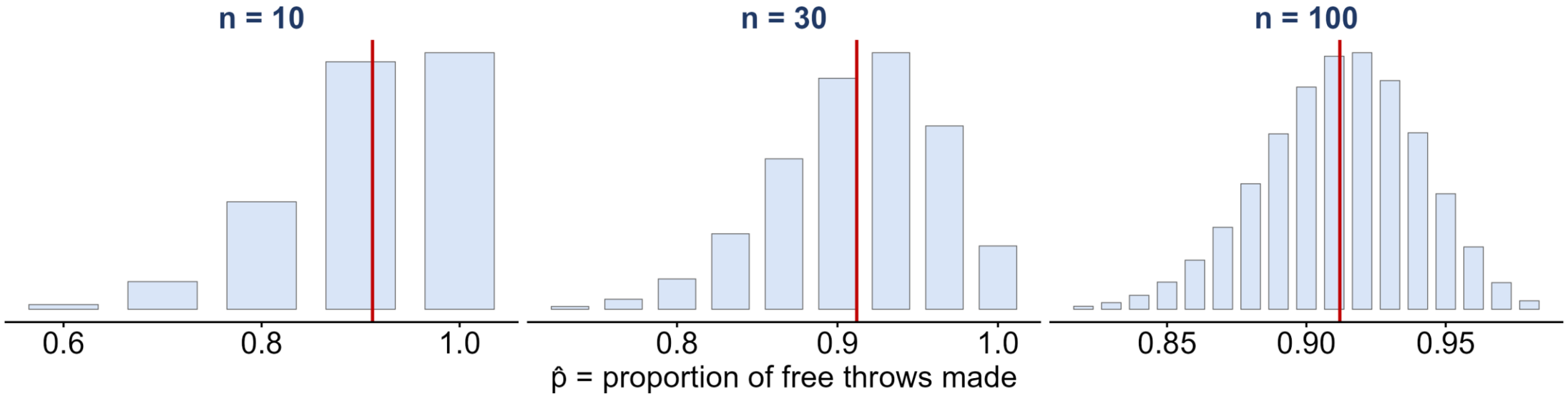
why it matters

**the empirical rule applies:  about 95% of samples give a statistic within 2 SE of its mean**

# The CLT for a proportion

Exact sampling distributions of $\hat{p}$, the proportion of free throws Stephen Curry makes; $p = 0.912$.

**For which n is $n(1 - p) > 5$? Which distributions look like a bell?**




**$n(1 - p) = 0.88,\ 2.64,\ 8.8$: only n = 100 passes, and only n = 100 is close to a bell**

# Example: free throws, again

Over a stretch of n = 100 free throws, let $\hat{p}$ be the proportion Curry makes.

**$p = 0.912$;  $n = 100$;  $n(1 - p) = 8.8 > 5$**

**(a) What are the mean and standard error of $\hat{p}$?**

**(b) About 95% of the time, $\hat{p}$ falls between what two values?**

**(a) mean 0.912;  SE = $\sqrt{(0.912 \times 0.088 / 100)} \approx 0.028$**

**(b) $0.912 \pm 2(0.028)$:  0.856 to 0.968**

# Example: a tournament's average

Goals per World Cup match: $X \sim \text{Poisson}(\lambda = 2.69)$. A tournament averages over 64 matches.

**$\mu = \lambda = 2.69$; $\sigma = \sqrt{2.69} \approx 1.64$; $n = 64$**

**(a) What is the standard error of a tournament's average?**
**(b) About 95% of tournaments average between what two values?**
**(c) How likely is a tournament that averages 3 or more goals per match?**

**(a) $1.64 / \sqrt{64} \approx 0.205$**
**(b) $2.69 \pm 2(0.205)$: 2.28 to 3.10 goals per match**
**(c) $z = (3 - 2.69) / 0.205 = 1.51$: $P(Z > 1.51) = 1 - 0.9345 = 0.0655$**

# From an estimate to an interval

point estimate

**a single statistic used to estimate a parameter: $\bar{x}$ for $\mu$, $\hat{p}$ for $p$**

confidence interval

**a range of plausible values for the parameter: estimate ± margin of error**

95% confidence

**the method captures the parameter in about 95% of all possible samples**

# Data: Spotify songs

Spotify and YouTube statistics for the top 10 songs of each of 2,079 popular artists, collected in February 2023.

**X = the length of a song, in minutes**

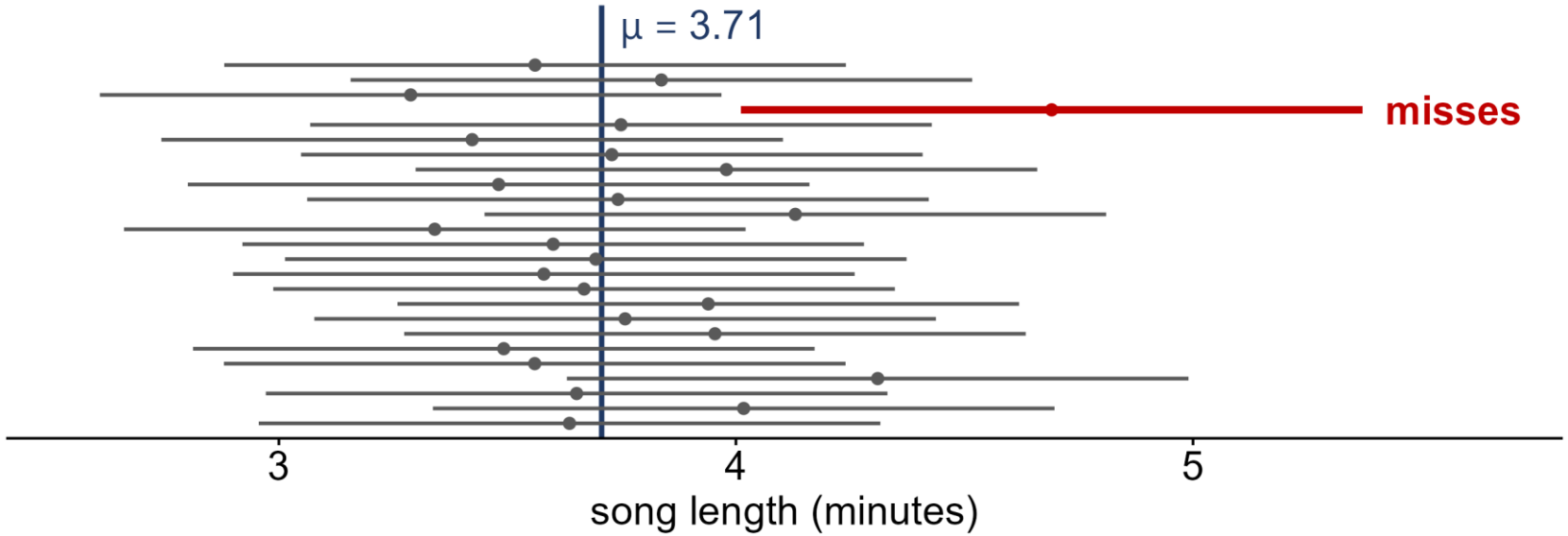
| TRACK | ARTIST | MINUTES |
|---|---|---|
| Feel Good Inc. | Gorillaz | 3.71 |
| Rhinestone Eyes | Gorillaz | 3.34 |
| New Gold (feat. Tame Impala and Bootie Brown) | Gorillaz | 3.59 |
| On Melancholy Hill | Gorillaz | 3.90 |
| Clint Eastwood | Gorillaz | 5.68 |
| … | … | … |

18,860 different songs (a song credited to several artists is counted once)

# How often does an interval capture μ?

25 random samples of n = 20 songs;  each line is $\bar{x} \pm 2(0.34)$, the margin of error.

**How many of the 25 intervals capture μ = 3.71?**




**24 of 25:  about 95% of intervals made this way capture μ**

# Example: an interval for mean song length

In practice we see only one sample: n = 20 songs (the bottom interval on the last slide).

**$\bar{x}$ = 3.64 min; $\sigma$ = 1.51 min; SE = 0.34 min**

**(a) What is the 95% confidence interval for $\mu$?**
**(b) Does it capture $\mu$ = 3.71?**
**(c) If we did not know $\mu$, what would "95% confident" mean?**

**(a) 3.64 ± 2(0.34): 2.96 to 4.32 min (b) Yes**
**(c) 95% of intervals built this way capture $\mu$; this one either does or does not**

# Confidence intervals for μ and for p

for a mean μ

**$\bar{x} \pm 2\sigma / \sqrt{n}$**

for a proportion p

**$\hat{p} \pm 2\sqrt{(\hat{p}(1 - \hat{p}) / n)}$: p is unknown, so the standard error uses $\hat{p}$**

when it works

**the CLT applies: $n \geq 30$ for $\bar{x}$; $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$ for $\hat{p}$**

# Data: teens and cell phones

A Pew Research Center survey asked 743 U.S. teens aged 13–17: "Are you losing focus in class by checking your cell phone?"

**X = 1 if a teen answers "sometimes" or "often", and 0 otherwise**

| RESPONSE | TEENS |
|---|---|
| Never | 289 |
| Rarely | 216 |
| Sometimes | 178 |
| Often | 60 |
| Total | 743 |

n = 743 teens

# Example: losing focus in class

Estimate p, the proportion of all U.S. teens who lose focus in class “sometimes” or “often”.

**178 + 60 = 238 of n = 743 teens**

**(a) What is $\hat{p}$? Are there more than 5 successes and 5 failures?**
**(b) What is the standard error of $\hat{p}$?**
**(c) What is the 95% confidence interval for p?**

**(a) $\hat{p} = 238 / 743 \approx 0.320$; 238 and 505 are both > 5**
**(b) $\sqrt{(0.320 \times 0.680 / 743)} \approx 0.017$**
**(c) $0.320 \pm 2(0.017)$: 0.286 to 0.354**

# What you should know

areas under the normal curve

**lower tail = table value;  upper tail = 1 − table value;  between = difference**

central limit theorem

**for large n, $\bar{x}$ and $\hat{p}$ are approximately normal, whatever the population**

confidence interval for $\mu$

**$\bar{x} \pm 2\sigma / \sqrt{n}$**

confidence interval for p

**$\hat{p} \pm 2\sqrt{\hat{p}(1 - \hat{p}) / n}$:  about 95% of intervals made this way capture p**